- The residuals (errors) are normally distributed and have constant variance.

By understanding these principles and limitations, you can leverage linear regression effectively to solve a variety of predictive modeling tasks.

Logistic Regression for Classification

Logistic regression is a fundamental machine learning algorithm widely used for binary classification problems. It predicts the probability of an observation belonging to one of two classes and is a cornerstone of predictive modeling in various fields such as healthcare, finance, and social sciences.

Unlike linear regression, which predicts a continuous outcome, logistic regression predicts a probability score bounded between 0 and 1 using the logistic (sigmoid) function. This score is then converted into binary outcomes based on a threshold, typically 0.5.

Why Use Logistic Regression?

Logistic regression is a popular choice for classification tasks because:

- **Simplicity:** It is easy to implement and interpret.
- **Efficiency:** Logistic regression is computationally lightweight and scales well with data.
- **Probabilistic Interpretation:** It provides probabilities for each class, allowing for nuanced decision-making.
- **Flexibility:** Logistic regression can be extended to handle multiclass classification tasks using techniques such as one-vs-rest or softmax regression.

Implementing Logistic Regression in Python

Step 1: Importing Required Libraries

To build a logistic regression model, you need libraries like Scikit-Learn and Pandas for data manipulation and modeling: from sklearn.linear_model import LogisticRegression from sklearn.model_selection import train_test_split from sklearn.metrics import accuracy_score